JACS
JOURNAL OF THE AMERICAN CHEMICAL SOCIETY
Article
Cite This: J. Am. Chem. Soc. 2019, 141, 7081−7089
pubs.acs.org/JACS

Chiral BINOL-Based Covalent Organic Frameworks for Enantioselective Sensing

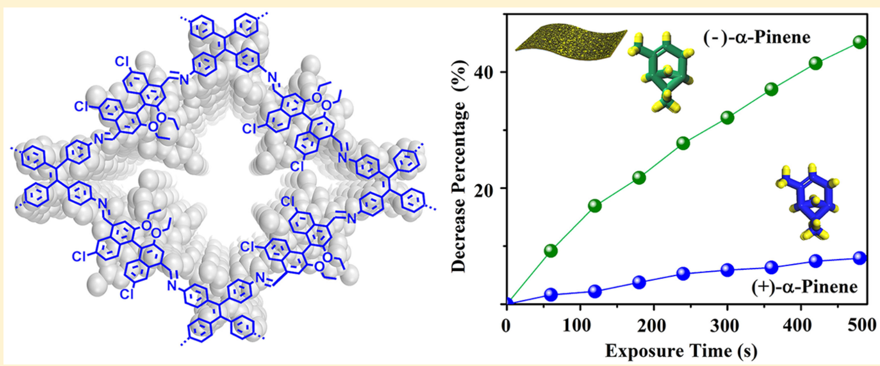
Xiaowei Wu,[†] Xing Han,[†] Qisong Xu,[‡] Yuhao Liu,[†] Chen Yuan,[†] Shi Yang,[†] Yan Liu,[†] Jianwen Jiang,[‡] and Yong Cui*,[†,§]

[†]School of Chemistry and Chemical Engineering and State Key Laboratory of Metal Matrix Composites, Shanghai Jiao Tong University, Shanghai 200240, China

[‡]Department of Chemical and Biomolecular Engineering, National University of Singapore, 4 Engineering Drive 4, 117576, Singapore

[§]Collaborative Innovation Center of Chemical Science and Engineering, Tianjin 300072, China

Supporting Information



ABSTRACT: Covalent organic frameworks (COFs) have emerged as a novel platform for material design and functional explorations, but it remains a challenge to synthetically functionalize targeted structures for task-specific applications. Optically pure 1,1′-bi-2-naphthol (BINOL) is one of the most important sources of chirality for organic synthesis and materials science, but it has not yet been used in construction of COFs for enantioselective processes. Here, by elaborately designing and choosing an enantiopure BINOL-based linear dialdehyde and a tris(4-aminophenyl)benzene derivative or tetrakis(4-aminophenyl)ethene as building blocks, two imine-linked chiral fluorescent COFs with a 2D layered hexagonal or tetragonal structure are prepared. The COF containing flexible tetraphenylethylene units can be readily exfoliated into ultrathin 2D nanosheets and electrospun to make free-standing nanofiber membrane. In both the solution and membrane systems, the fluorescence of COF nanosheets can be effectively quenched by chiral odor vapors via supramolecular interactions with the immobilized BINOL moieties, leading to remarkable chiral vapor sensors. Compared to the BINOL-based homogeneous and membrane systems, the COF nanosheets exhibited greatly enhanced sensitivity and enantioselectivity owing to the confinement effect and the conformational rigidity of the sensing BINOL groups in the framework. The ability to place such a useful BINOL chiral auxiliary inside open channels of COFs capable of amplifying chiral discrimination of the analytes represents a major step toward the rational synthesis of porous molecular materials for more chirality applications.

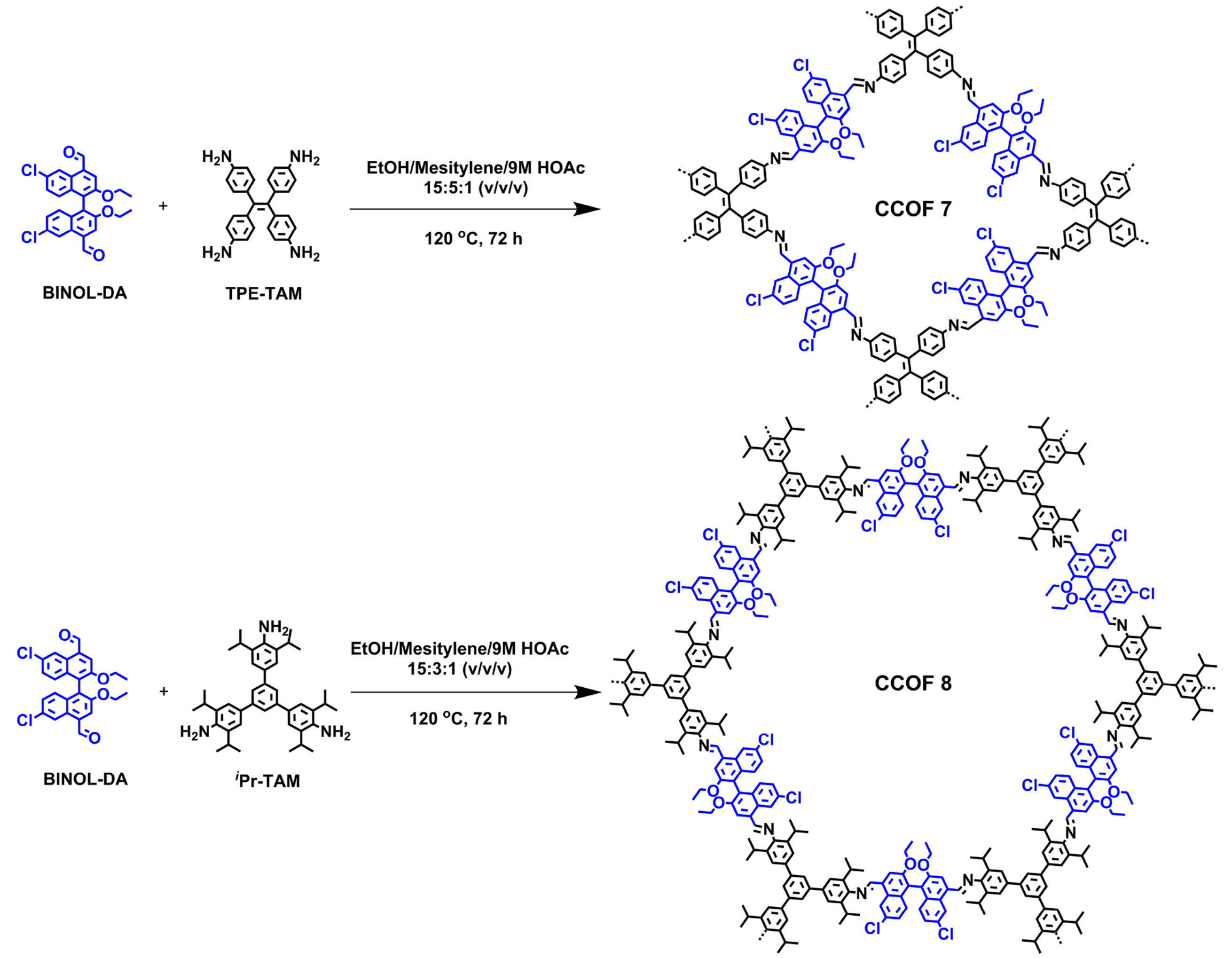
INTRODUCTION

Covalent organic frameworks (COFs) are an emerging class of porous crystalline polymers constructed from organic monomers linked by covalent bonds[1] with potential applications in diverse areas such as gas storage[2] and separation,[3] optoelectronics,[4] energy storage,[5] sensing,[6] and catalysis.[7] COFs are advantageous over other inorganic materials in their versatility as many features including their chemical bonding types, constituent elements, and functional groups, pore sizes and shapes, may be readily fine-tuned. In particular, judicious choices of building blocks can lead to porous chiral COFs (CCOFs) that can be utilized for enantioseletive progresses that cannot be achieved with traditional porous inorganic materials.[8−12] Well-defined molecular chiral catalysts derived from TADDOL,[9] metallosalen,[10] and L-proline[11] have been incorporated into COFs to prepare heterogeneous catalysts with activities and enantioselectivities rivaling those of homogeneous counterparts. Progress has also been made in incorporating chiral hydroxyl functionalities into COF structures to afford stationary phases for chromatographic separation of enantiomers.[12] However, it remains a challenge to introduce functional groups in CCOFs for other important applications such as chiral optics and sensing. It should be

Received: February 25, 2019
Published: April 11, 2019

Downloaded via UNIV SCIENCE AND TECHNOLOGY CHINA on August 2, 2025 at 08:45:42 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

Scheme 1. Synthesis of the CCOFs



noted that chiral discrimination of vapors plays an important role in olfactory perception of biological systems and its realization by artificial sensors has been an intriguing challenge.[13] Here we demonstrate that fluorescent CCOFs with suitable recognition sites can allow chirality sensing of terpene flavor molecules, which are a large and diverse class of naturally occurring organic chemicals and are widely used as industrially relevant chemicals.[14]

Optically pure 1,1′-bi-2-naphthol (BINOL) is one of the most important sources of chirality for organic synthesis and materials science, and its versatility and availability make it an ideal platform for asymmetric catalysis, chiral recognition, and optics.[15] Using BINOL as the chirality source as well as the core fluorophore, structurally diverse enantioselective fluorescence sensors including organic oligomers, dentrimers, and polymers have been constructed for chiral compounds with hydroxyl, amino, and carboxylate groups that are capable of forming hydrogen bonds with the two BINOL hydroxyl groups.[16] Recently, BINOL has been incorporated into metal−organic materials to generate chiral pockets for enhanced enantioselectivity in luminescence quenching of amino alcohol relative to their building blocks.[17] Despite the importance of the BINOL, it has not yet been used in the crystallization of COFs. In this work, axially chiral linker 6,6′-dichloro-2,2′-diethoxy-1,1′-binaphthyl-4,4′-dialdehyde (**BINOL-DA**), which has the dialdehyde primary functionality and orthogonal chiral 2,2′-diethoxy secondary functionality, was designed and synthesized to crystallize CCOFs. By imine condensations of this chiral linker and tetrakis(4-aminophenyl)ethene (**TPE-TAM**) or 1,3,5-tris(4-amino-3,5-diisopropylphenyl)benzene (**^{i}Pr-TAM**), two fluorescent CCOFs with a 2D layered tetragonal or hexagonal structure were prepared (Scheme 1). The COF containing TPE building units can be exfoliated into ultrathin 2D nanosheets (NSs) for fabricating free-standing nanofiber membrane by electrospinning. The resulting CCOF NSs are remarkably sensitive to vapor quencher in both the solution and membrane with much enhanced enantioselectivity compared to the corresponding BINOL-based system. The difference is presumably ascribed to the channel confinement effect and conformational rigidity of the sensing BINOL sites in the COF structure.

■ RESULTS AND DISCUSSION

Synthesis and Characterization. CCOFs 7 and 8 were synthesized by heating enantiopure (*R*)-**BINOL-DA** (46.7 mg, 0.1 mmol) with **TPE-TAM** (19.6 mg, 0.05 mmol) or **^{i}Pr-TAM** (40.4 mg, 0.067 mmol) in different solvents in the presence of acetic acid at 120 °C for 3 days, which afforded yellow crystalline solids in 62 and 77% yields, respectively (Scheme 1). Both COFs are stable in common organic solvents. In the FT-IR spectra, the characteristic C═O stretching bands

DOI: 10.1021/jacs.9b02153
J. Am. Chem. Soc. 2019, 141, 7081−7089

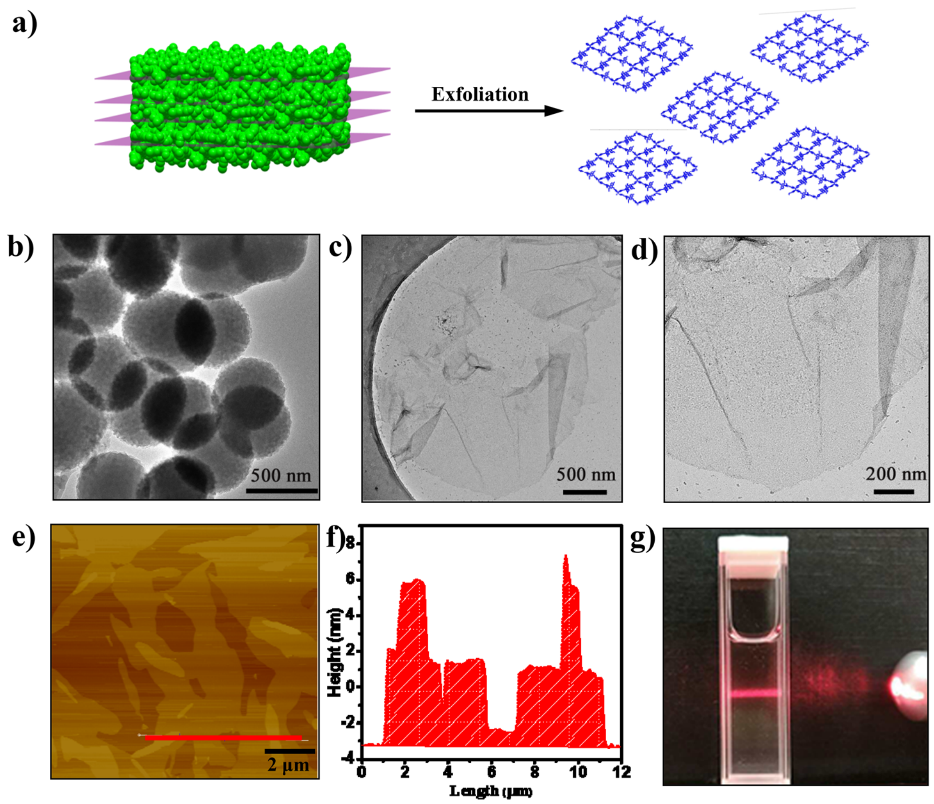


Figure 1. (a) Schematic diagram for the exfoliation of CCOF **7** to ultrathin 2D NS. TEM images of (b) the bulk **7** and (c, d) **7-NS**. (e) AFM image of the 7-NS. (f) Height of AFM image for the selective area. (g) Photograph of the Tyndall effect of the NS suspension in MeOH.

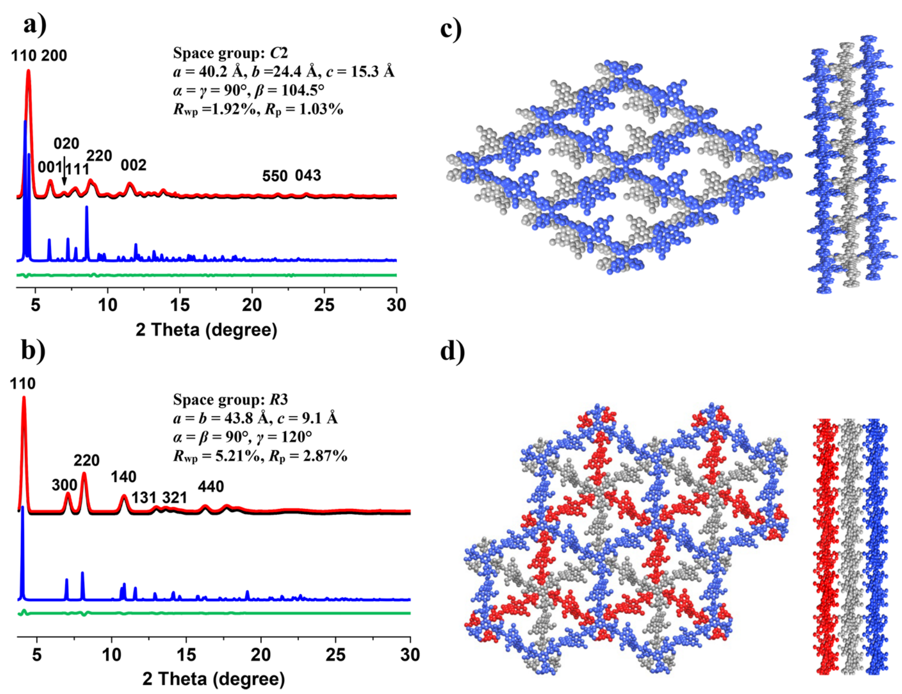


Figure 2. PXRD patterns of CCOF **7** (a) and **8** (b) with the experimental profiles in black, Pawley-refined profiles in red, calculated profiles in blue, and the differences between the experimental and refined PXRD patterns in dark cyan. Top (left) and side (right) views of the corresponding refined 2D crystal structures of (c) **7** and (d) **8**.

($\sim$1700 cm^{-1}) almost disappeared, indicative of the consumption of the aldehydes (Figure S1). Strong stretching vibration bands attributed to the new generation of C═N linkages were observed at $\sim$1621 cm^{-1} (**7**) and $\sim$1620 cm^{-1} (**8**). In the ^{13}C CP-MAS NMR spectra, isopropyl group in CCOF **8** signals appeared at 23 and 31 ppm; the ethoxy group in **BINOL-DA** showed signals at 65 and 15 ppm for both **7** and **8**. The characteristic signals due to C═N bonds were

DOI: 10.1021/jacs.9b02153
J. Am. Chem. Soc. 2019, 141, 7081−7089

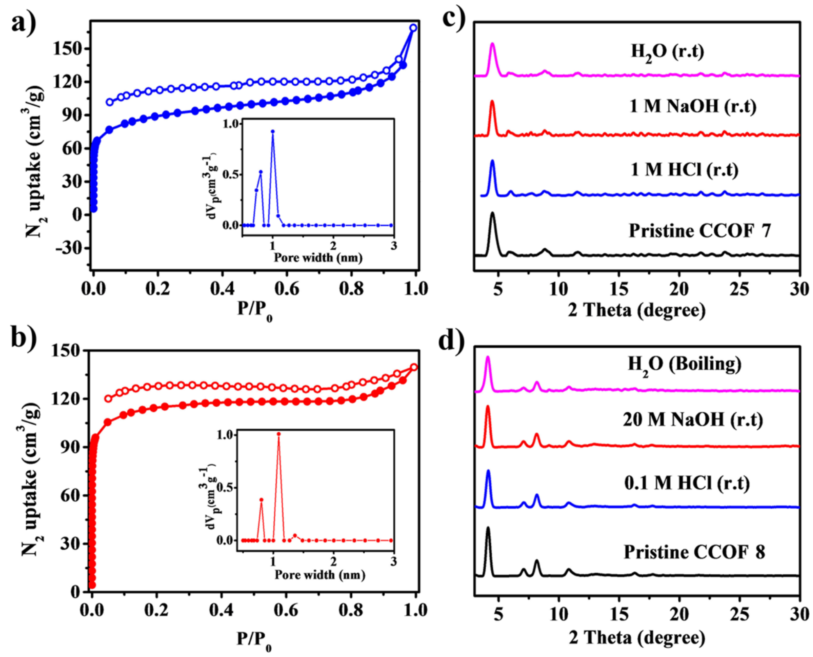


Figure 3. N_2 adsorption–desorption isotherms and pore size distribution profiles (insert) for CCOFs **7** (a) and **8** (b). PXRD patterns of (c) **7** and (d) **8** after 7 days treatment under different conditions.

observed at 157 ppm (Figure S2). The aldehyde carbon peaks were barely present. The chemical shifts of other fragments are consistent with those of the monomers. Thermal gravimetric analysis revealed that those COFs exhibited no weight loss under N_2 to 380 °C (Figure S4). Sphere-like morphology were observed from the SEM and TEM images for the two CCOFs (Figure S5a). Circular dichroism (CD) spectra of these CCOFs made from *R* and *S* enantiomers of the BINOL monomers are mirror images of each other, which is indicative of their enantiomeric nature (Figure S3).

Due to the presence of twisted TPE species in CCOF **7** that may prevent effective layer packing, this CCOF can be easily exfoliated into nanosheets (denoted as **7**-NS) through solvent-assisted liquid sonication[18] (section 2.3 in the Supporting Information). TEM images revealed that after exfoliation the spherelike bulk CCOF **7** was exfoliated into sheetlike layered structure of the 2D nanosheets (Figure 1b–d). AFM characterization (Figure 1e,f) revealed that the thickness of nanosheets is 3.5–4.0 nm, corresponding to 5–6 interlayers in the CCOF. The suspension of nanosheets in MeOH exhibits a typical Tyndall effect (Figure 1g), indicating the colloidal feature of the free-standing and homogeneous ultrathin nanosheets. More characterizations including FT-IR, CD and TGA also demonstrated the successful preparation of 2D nanosheets (Figures S1–S6). Attempts to exfoliate the bulk CCOF **8** into nanosheets through the same approach have failed thus far.

Crystal Structure. The crystalline structures of the CCOFs were determined by powder X-ray diffraction (PXRD) analysis with Cu Kα radiation. As revealed from PXRD analyses, CCOF **7** shows high crystallinity, exhibiting the first intense peak at a low angle 4.5° (2θ), which corresponds to the (110) reflection plane, along with minor peaks at 5.9, 7.6, 7.9, 8.8, 11.6, 21.6, and 23.3°, attributed to the (001), (020), (111), (220), (002), (550) and (043) reflection planes, respectively (Figure 2a). For CCOF **8**, the first and most intense peak corresponding to the (110) reflection plane appears at 4.1°, with other minor peaks at 7.0, 8.1, 10.8, 12.9, 14.1, and 16.2°, attributed to the (300), (220), (140), (131), (321), and (440) reflection planes, respectively (Figure 2b). In order to elucidate the structure of these COFs and to calculate the unit cell parameters, two types of possible 2D structures were generated for each of them, that is, eclipsed or slipped stacking (AA) and staggered stacking (AB and ABC) models were built and optimized by the Materials Studio Forcite molecular dynamics module method. The experimental PXRD pattern for **7** matched well with the simulated patterns of the slipped stacking (AA) model in the monoclinic system *C*2 space group. The models proposed for PXRD patterns of **8** agreed well with the simulated pattern generated from the staggered stacking (ABC) of the 2D layers in the trigonal *R*3 space group. Pawley refinements gave optimized parameters as follows: $a = 40.2$ Å, $b = 24.4$ Å, $c = 15.3$ Å, $\alpha = \gamma = 90°$, $\beta = 104.5°$ for **7**; $a = b = 43.8$ Å, $c = 9.1$ Å, $\alpha = \beta = 90°$, $\gamma = 120°$ for **8**. These provided good agreement factors: $R_{wp} = 1.92\%$ and $R_p = 1.03\%$ for **7**; $R_{wp} = 5.21\%$ and $R_p = 2.87\%$ for **8**. PXRD patterns were also calculated for the two CCOFs on the other structures, but the calculated PXRD patterns did not match well the experimental patterns (Figures S32 and S33).

The porosity of the CCOFs was examined by measuring N_2 sorption isotherms at 77 K on the activated samples. The adsorption curves of them exhibited type-I isotherm (Figure 3), a characteristic of microporous materials. The Brunauer–Emmett–Teller (BET) surface areas of the bulk samples of **7** and **8** were calculated to be 281 and 346 $m^2 \cdot g^{-1}$. The total pore volumes were estimated to be 0.37 and 0.53 $cm^3 g^{-1}$ at $P/P_0 = 0.99$, respectively. The nonlocal density functional theory (NLDFT) gave rise to a narrow pore size distribution with an average pore width 0.81 and 1.1 nm for **7** and 0.80 and 1.0 nm for **8**, corresponding to their simulated values. In contrast, the BET surface area of **7**-NS was decreased to 98 $m^2 \cdot g^{-1}$ (Figure S9). This is probably due to after exfoliation the long-range pore structure of the CCOF is disturbed and that only shallow

DOI: 10.1021/jacs.9b02153
J. Am. Chem. Soc. 2019, 141, 7081–7089

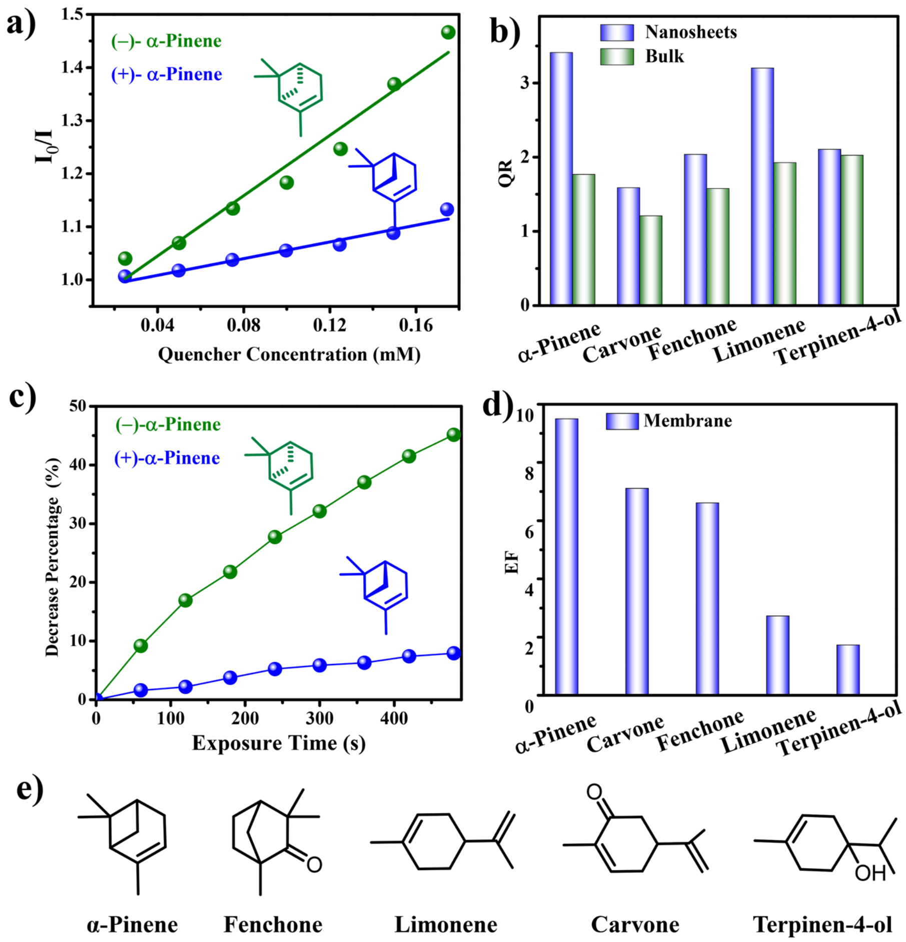


Figure 4. (a) Stern−Völmer plots of 7-NS upon titration of α-pinene in MeCN. (b) Enantioselective quenching ratio for several terpenes. (c) Decrease percentage upon exposure to α-pinene for 7@PVDF (d) Enantioselective fluorescence decrease ratio for several vapors. (e) Chemical structures of terpenes used in the study.

pores are accessible for N_2 sorption, thereby leading to only moderate surface area value.

Chemical Stability. The chemical stability of the two CCOFs was examined by PXRD and N_2 sorption isotherms after 7 days of treatment in water, HCl (aqueous) and NaOH (aqueous). It was found that 7 was stable in 1 M HCl (aqueous), 1 M NaOH (aqueous), and water (r.t.), whereas **8** was stable in 0.1 M HCl (aqueous), 20 M NaOH (r.t.), and boiling water, as evidenced by the almost unchanged PXRD patterns (Figure 3c,d), BET surface areas (Figure S9), and residue weight percentage (Figure S10). Therefore, CCOF **8** showed improved alkali resistance and antihydrolysis capability relative to 7, consistent with that the pendant bulky isopropyl groups can protect imine linkage and hydrolytically susceptible backbones through kinetic blocking.[19]

Enantioselective Sensing. Chiral discrimination of vapors is of critical importance in many areas of analytical chemistry and biotechnology. It has been an intriguing challenge to realize special artificial odor sensors for chiral discrimination in drug analysis and environmental monitoring.[13] As expected, CCOF 7 is highly fluorescent, with an emission maximum at 380 nm in solid state, different from **BINOL-DA** and **TPE-TAM**, which give emission maxima at 483 and 494 nm, respectively. Moreover, with different excitation wavelengths, varied emission spectra of the CCOF 7 at around 335, 420, 535, 576, and 630 nm are acquired (Figure S7a), indicating the presence of diverse emission specie.[20]

It is likely that the fluorescence of the CCOF **7** is associated with both **BINOL-DA** and **TPE-TAM**. 7-NS is also highly fluorescent in acetonitrile with an emission maximum at 380 nm and an emission lifetime of <2 ns. The emission was blue-shifted by about 40 nm compared to that of bulk powder (Figures S19 and S25), consistent with the reduced interlayer $\pi-\pi$ interaction in nanosheets. The presence of large chiral pores and available ethoxyl groups in this BINOL CCOF prompted the exploration of enantioselective fluorescence recognition and sensing. 7-NS was examined for fluorescence recognition and sensing of chiral vapors including α-pinene, limonene, fenchone, carvone, and terpinen-4-ol.

7-NS was dispersed in acetonitrile to prepare a stock solution with the BINOL unit at a concentration of 10 μM. Aliquots containing different amounts of the (−) -and (+) -enantiomers of the substrates were added to acetonitrile suspensions of 7-NS (2 mL), and the fluorescence signals of the NS suspensions in the presence of different amounts of substrates were measured. The results for titrations of 7-NS with α-pinene are shown in Figure 4. When 7-NS were treated with the two enantiomers of (α)-pinene, the emission at 380 nm was both decreased, but the rate of quenching caused by the (−)- enantiomer was faster than that with the (+) enantiomer, implying enantioselectivity in the fluorescent

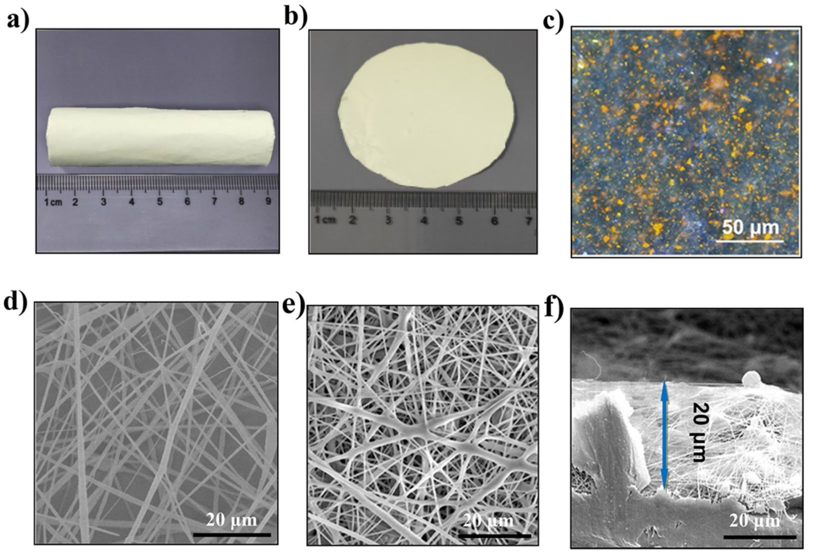


Figure 5. Optic photos of (a) crimped and (b) flat 7@PVDF membrane. (c) The fluorescence photo of the 7@PVDF membrane irradiated by UV light. SEM images of (d) the pure PVDF and (e) 7@PVDF membranes. (f) Cross-sectional view of SEM images of 7@PVDF membrane.

recognition. In accordance with the linear Stern−Völmer equation, the measured absorbance $[I_0/I]$ varied as a function of [M] in a linear relationship, indicating 1:1 stoichiometry of the interaction between the BINOL unit of 7-NS and (α)-pinene. The K_{sv} constants were calculated to be 1348 and 395 M^{-1} with the (−) -and (+) -enantiomers, respectively, affording the quenching ratio [QR = $K_{sv}(-)/K_{sv}(+)$] of 3.41 (Figure 4a). The opposite trend in stereoselectivity was observed for quenching (S)-7-NS with (α)-pinene, for which the QR value was 3.49, further demonstrating the stereoselectivity in the fluorescent recognition (Figure S24). After titration, the quantum yield of 7-NS slightly decreased from 7.6 to 6.3%, and the fluorescence lifetimes changed from 1.81 and 1.74 ns, respectively, probably as a result the nonradiative decay[26a] (Figures S30 and S31). Besides, 7-NS are also enantioselective toward limonene, fenchone, terpinen-4-ol, and carvone, with the QR values ranging from 1.20 to 3.41(Figure 4b).

In practical applications, it is beneficial to process the COF materials into films, filters, or shape-bodies to avoid the problems such as clogging or recycling issues. COF membranes have been prepared by growing/depositing COFs on porous substrates or incorporating COFs powders/nanosheets into polymers (mixed matrix membranes).[21] Electrospinning is a common and facile method to fabricate fibers, and literature reports include the preparation of electrospun fibers with MOFs, zeolites, carbon materials, and silicates as the fillers.[22] However, there are only two examples on processing COFs membranes with electrospinning.[23]

We investigated the utilization of electrospinning for immobilization of 7-NS in nanofibers to make CCOF membranes. A small portion of PVDF (polyvinylidene fluoride) was first added to the 7-NS dispersion in DMF and acetone (1:1 weight ratio) to form a polymer coating on the NSs, and then the rest of the polymer was added to get the electrospinning solution. We performed different loading of 7-NS in PVDF matrix ranging from 5 to 10 wt % for electrospinning and increasing the loading (more than 10 wt %) led to unsuccessful electrospinning. By adjusting the electrospinning parameters such as applied electric voltage and flow rate of the solution, the 7-NS/PVDF solution was electrospun onto aluminum foil substrates to produce free-standing nanofiber membranes 7@PVDF (section 2.4 in Supporting Information). SEM images showed that 7-NS were well-dispersed in the PVDF nanofibers that have an average diameter of 200−300 nm (Figure 5e). Cross-sectional view of the SEM image showed the thickness of membrane is about 20 μm (Figure 5f), which may facilitate the fast diffusion and transport of analytes. Fluorescence photography demonstrated the photoluminescence property of membrane and also indicated a homogeneous dispersion of 7-NS within the PVDF matrix (Figure 5c).

7@PVDF membrane (10 wt %) displayed a maxmum emission at 380 nm, whereas the pure PVDF membrane showed no fluorescence. The fluorescence spectra of the membrane were monitored upon exposure to the vapors of chiral molecules for various durations. When membrane was exposed to α-pinene, the emission at 380 nm was decreased by both the (−)- and (+)-enantiomers, but the rate of quenching caused by (−) was faster than that caused by (+) enantiomer (Figure 4c). The decrease percentage was estimated using the formula $(I_0 - I)/I_0 \times 100\%$, where I_0 is the maximum fluorescence intensity before exposure to the analyte, and I is the maximum intensity after exposure to the analyte. The decrease percentage of the (−)- and (+)-enantiomes were determined as 45 and 8% after 480 s, respectively (Figure 4c). The enantioselective fluorescence decrease ratio was calculated as the formula [EF = $(I_0/I_R - 1)/(I_0/I_S - 1)$] of 9.5. Importantly, the 7@PVDF membrane was also enantioselective toward a range of chiral vapors, including limonene, fenchone, terpinen-4-ol, and carvone (Figure 4d). The EF values range from 1.7 to 9.5, comparable with those of luminescent sensors based on metal−organic material, molecular, and polymer sensors for small chiral molecules.[17a,24] Control experiments indicated that the **BINOL-DA**/PVDF membranes (section 2.4 in the Supporting Information) showed almost no fluorescence changes with (−)- or (+)-α-pinene (Figure S11), indicating no enantioselectivity. Moreover, after each measurement, 7@PVDF membrane could be regenerated by heating at 100 °C in vacuum for about 1 h and

DOI: 10.1021/jacs.9b02153
J. Am. Chem. Soc. 2019, 141, 7081−7089

directly reused for the next cycle of sensing without significant loss of enantioselectivity (EF = 9.5, 9.4, and 9.2. for runs 1−3 in sensing α-pinene, respectively, Figure S17). After exposure to α-pinene, the quantum yield of 7@PVDF slightly decreased from 1.5 to 1.1%, and the fluorescence lifetime varied from 1.42 to 1.36 ns. After three cycles of sensing, the structure of 7@PVDF membrane remain almost unchanged, as indicated by the SEM image (Figure S5b)

The observed fluorescence quenching in both 7-NS and 7@PVDF may be ascribed to static enhancement upon formation of a CCOF−vapor adduct, which can induce changes in the structure of emitting species such as a change and/or rigidification of the conformation and excimer formation.[25] The static nature of the complexation is indicated by consistent fluorescence lifetimes of 7-NS and 7@PVDF before and after titration with/exposure to (−)-α-pinene (1.81 vs 1.74 ns and 1.42 vs 1.36 ns, respectively) (Figure S30). In the 7-NS solution system, due to the short excited-state lifetime (<2 ns), low fluorophore concentration (10 μM), and quencher concentration (0−150 μM), the collision probability during the excited state lifetime of the fluorophore was very low, so the collisional quenching was negligible. Additionally, there is no overlap between the absorption (emission) spectra of CCOF 7 and the emission (absorption) spectra of (−)-α-pinene (Figure S7d), indicating no obvious electron or energy transfer between them.[26]

Control experiments showed that the bulk sample of CCOF 7, under identical conditions, exhibited much lower enantioselective performance than 7-NS (Figure 4b). This feature may be attributed to the NSs have more exposed external surface and accessible active binding sites, allowing sufficient contact and interaction with the analyte.[6d,18] Control experiments also indicated that the monomer **BINOL-DA** in both the solution and composited PVDF membrane displayed no fluorescence changes with (−)- or (+)-α-pinene and no stereoselectivity (Figures S11 and S18). It is thus likely that the steric confinement of the CCOF channels can create a discriminating chiral environment for α-pinene, affording a high enantioselectivity.

Molecular simulations were performed to estimate the binding energies of the analytes in CCOF 7. As shown in Figure S35, both enantiomers of α-pinene are found to preferentially locate in the microenvironments created by two rigid BINOL cores and their ethoxyl groups. However, the structural orientation of the (−)-enantiomer at the binding site is clearly different from that of the (+)-enantiomer. For the (−)-enantiomer, the double bonds and aliphatic groups are observed in the proximity of the BINOL units, leading to stronger host−guest interaction compared with the (+)-enantiomer. The binding energy of (−)-enantiomer is more negative than that of the (+)-enantiomer (−21.92 vs −18.63 kcal/mol), indicating stronger binding affinity between the (−)-enantiomer and the COF (Table S1). A similar behavior was observed for limonene (Figure S36). Therefore, the different binding affinities between two enantiomers give rise to chiral discrimination.

Various procedures for sensing enantiomers of terpenes molecules have been explored and those of note are mostly based on electrochemical methods (for example organic field effect transistors), but they are costly and time-consuming or unsatisfactory enantioselective (Table S2).[27] Chiral recognition of vapors using optical methods has been a challenge because volatile small chiral molecules can generally induce only a small structure change in chromophore cores, which is hard to detect.[27a] In contrast, COFs have been extensively explored as fluorescence sensors for a variety of important compounds such as explosives,[6a,b] volatile organic compounds,[6c] biomolecules,[6d] and humidity;[6e] however, enantioselective recognition and detection of chiral compounds have not yet been achieved so far. By taking advantages of the intense fluorescence of the BINOL skeleton, we demonstrated for the first time that COFs can be used as optical gas sensor for terpenes with high sensitivity and enantioselectivity. The present solid-state enantioselective sensors hold the potential to carry out online enantiomer discrimination, with exciting perspectives in process monitoring.

■ CONCLUSION

In summary, by co-condensation of an enantiopure BINOL-based linear dialdehyde and a tri- or tetraamine, we have synthesized two fluorescent CCOFs with a layered hexagonal or tetragonal network. The CCOF containing flexible TPE units can be easily exfoliated into ultrathin nanosheets and electrospun into free-standing PVDF nanofiber membranes. As a proof-of-concept application, the obtained ultrathin CCOF 7-NSs were used as a novel fluorescence-sensing platform for the detection of chiral odor vapors in both the solution and membrane. The CCOF NSs exhibited much superior selectivity and enantiosensitivity to the corresponding BINOL-based sensing systems, probably as a consequence of steric confinement of the CCOF channels and conformational rigidity of the immobilized BINOL groups. The amplified chiral discrimination of the analytes afforded by the BINOL-based COF should be utilized to design functional CCOFs for other targeted chirality applications such as heterogeneous asymmetric catalysis, enantioseparations, chiral optics, and so forth.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/jacs.9b02153.

Experimental procedures and characterization data (PDF)

Crystallographic information files for CCOF 7 and CCOF **8** (CIF, CIF)

■ AUTHOR INFORMATION

Corresponding Author

*E-mail: yongcui@sjtu.edu.cn.

ORCID

Xiaowei Wu: 0000-0003-2141-3448
Xing Han: 0000-0002-7921-8197
Qisong Xu: 0000-0002-1337-543X
Yuhao Liu: 0000-0002-7222-4527
Chen Yuan: 0000-0003-0334-2594
Yan Liu: 0000-0002-7560-519X
Jianwen Jiang: 0000-0003-1310-9024
Yong Cui: 0000-0003-1977-0470

Notes

The authors declare no competing financial interest.

DOI: 10.1021/jacs.9b02153
J. Am. Chem. Soc. 2019, 141, 7081−7089

■ ACKNOWLEDGMENTS

We acknowledged the technical support in electrospinning from Jin Chen, Kunming Shi, and Shijie Hou. This work was financially supported by the National Natural Science Foundation of China (Grants 21431004, 21620102001, 21875136, and 91856204), the National Key Basic Research Program of China (Grant 2016YFA0203400), and Key Project of Basic Research of Shanghai (17JC1403100 and 18JC1413200).

■ REFERENCES

(1) (a) Côté, A. P.; Benin, A. I.; Ockwig, N. W.; O'Keeffe, M.; Matzger, A. J.; Yaghi, O. M. Porous, Crystalline, Covalent Organic Frameworks. *Science* **2005**, *310*, 1166. (b) Kandambeth, S.; Mallick, A.; Lukose, B.; Mane, M. V.; Heine, T.; Banerjee, R. Construction of Crystalline 2D Covalent Organic Frameworks with Remarkable Chemical (Acid/Base) Stability via a Combined Reversible and Irreversible Route. *J. Am. Chem. Soc.* **2012**, *134*, 19524. (c) Jin, E.; Asada, M.; Xu, Q.; Dalapati, S.; Addicoat, M. A.; Brady, M. A.; Xu, H.; Nakamura, T.; Heine, T.; Chen, Q.; Jiang, D. Two-Dimensional sp^2 Carbon−Conjugated Covalent Organic Frameworks. *Science* **2017**, *357*, 673. (d) Li, H.; Chavez, A. D.; Li, H.; Li, H.; Dichtel, W. R.; Bredas, J.-L. Nucleation and Growth of Covalent Organic Frameworks from Solution: The Example of COF-5. *J. Am. Chem. Soc.* **2017**, *139*, 16310.

(2) (a) Han, S. S.; Furukawa, H.; Yaghi, O. M.; Goddard, W. A. Covalent Organic Frameworks as Exceptional Hydrogen Storage Materials. *J. Am. Chem. Soc.* **2008**, *130*, 11580. (b) Tilford, R. W.; Mugavero, S. J., 3rd; Pellechia, P. J.; Lavigne, J. J. Tailoring Microporosity in Covalent Organic Frameworks. *Adv. Mater.* **2008**, *20*, 2741. (c) Yang, Y.; Faheem, M.; Wang, L.; Meng, Q.; Sha, H.; Yang, N.; Yuan, Y.; Zhu, G. Surface Pore Engineering of Covalent Organic Frameworks for Ammonia Capture through Synergistic Multivariate and Open Metal Site Approaches. *ACS Cent. Sci.* **2018**, *4*, 748.

(3) (a) Ji, W.; Xiao, L.; Ling, Y.; Ching, C.; Matsumoto, M.; Bisbey, R. P.; Helbling, D. E.; Dichtel, W. R. Removal of GenX and Perfluorinated Alkyl Substances from Water by Amine-Functionalized Covalent Organic Frameworks. *J. Am. Chem. Soc.* **2018**, *140*, 12677. (b) Shinde, D. B.; Sheng, G.; Li, X.; Ostwal, M.; Emwas, A.-H.; Huang, K.-W.; Lai, Z. Crystalline 2D Covalent Organic Framework Membranes for High-Flux Organic Solvent Nanofiltration. *J. Am. Chem. Soc.* **2018**, *140*, 14342. (c) Ning, G.; Chen, Z.; Gao, Q.; Tang, W.; Chen, Z.; Liu, C.; Tian, B.; Li, X.; Loh, K. Salicylideneanilines-Based Covalent Organic Frameworks as Chemoselective Molecular Sieves. *J. Am. Chem. Soc.* **2017**, *139*, 8897. (d) Zhang, S.; Zheng, Y.; An, H.; Aguila, B.; Yang, C.-X.; Dong, Y.; Xie, W.; Cheng, P.; Zhang, Z.; Chen, Y.; Ma, S. Covalent Organic Frameworks with Chirality Enriched by Biomolecules for Efficient Chiral Separation. *Angew. Chem., Int. Ed.* **2018**, *57*, 16754.

(4) (a) Bertrand, G. H. V.; Michaelis, V. K.; Ong, T.; Griffin, R. G.; Dincǎ, M. Thiophene-based Covalent Organic Frameworks. *Proc. Natl. Acad. Sci. U. S. A.* **2013**, *110*, 4923. (b) Bessinger, D.; Ascherl, L.; Auras, F.; Bein, T. Spectrally Switchable Photodetection with Near-Infrared-Absorbing Covalent Organic Frameworks. *J. Am. Chem. Soc.* **2017**, *139*, 12035. (c) Ding, H.; Li, J.; Xie, G.; Lin, G.; Chen, R.; Peng, Z.; Yang, C.; Wang, B.; Sun, J.; Wang, C. An AIEgen-based 3D Covalent Organic Framework for White Light-Emitting Diodes. *Nat. Commun.* **2018**, *9*, 5234. (d) Haldar, S.; Chakraborty, D.; Roy, B.; Banappanavar, G.; Rinku, K.; Mullangi, D.; Hazra, P.; Kabra, D.; Vaidhyanathan, R. Anthracene-Resorcinol Derived Covalent Organic Framework as Flexible White Light Emitter. *J. Am. Chem. Soc.* **2018**, *140*, 13367.

(5) (a) Halder, A.; Ghosh, M.; Khayum M, A.; Bera, S.; Addicoat, M.; Sasmal, H. S.; Karak, S.; Kurungot, S.; Banerjee, R. Interlayer Hydrogen-Bonded Covalent Organic Frameworks as High-Performance Supercapacitors. *J. Am. Chem. Soc.* **2018**, *140*, 10941. (b) Wei, H.; Ning, J.; Cao, X.; Li, X.; Hao, L. Benzotrithiophene-Based Covalent Organic Frameworks: Construction and Structure Transformation under Ionothermal Condition. *J. Am. Chem. Soc.* **2018**, *140*, 11618. (c) Du, Y.; Yang, H.; Whiteley, J. M.; Wan, S.; Jin, Y.; Lee, S.; Zhang, W. Ionic Covalent Organic Frameworks with Spiroborate Linkage. *Angew. Chem., Int. Ed.* **2016**, *55*, 1737. (d) Lv, J.; Tan, Y.-X.; Xie, J.; Yang, R.; Yu, M.; Sun, S.; Li, M.-D.; Yuan, D.; Wang, Y. Direct Solar-to-Electrochemical Energy Storage in a Functionalized Covalent Organic Framework. *Angew. Chem., Int. Ed.* **2018**, *57*, 12716.

(6) (a) Dalapati, S.; Jin, S.; Gao, J.; Xu, Y.; Nagai, A.; Jiang, D. An Azine-Linked Covalent Organic Framework. *J. Am. Chem. Soc.* **2013**, *135*, 17310. (b) Rao, M. R.; Fang, Y.; DeFeyter, S.; Perepichka, D. F. Conjugated Covalent Organic Frameworks via Michael Addition−Elimination. *J. Am. Chem. Soc.* **2017**, *139*, 2421. (c) Das, G.; Biswal, B. P.; Kandambeth, S.; Venkatesh, V.; Kaur, G.; Addicoat, M.; Heine, T.; Verma, S.; Banerjee, R. Chemical Sensing in Two Dimensional Porous Covalent Organic Nanosheets. *Chem. Sci.* **2015**, *6*, 3931. (d) Peng, Y.; Huang, Y.; Zhu, Y.; Chen, B.; Wang, L.; Lai, Z.; Zhang, Z.; Zhao, M.; Tan, C.; Yang, N.; Shao, F.; Han, Y.; Zhang, H. Ultrathin Two-Dimensional Covalent Organic Framework Nanosheets: Preparation and Application in Highly Sensitive and Selective DNA Detection. *J. Am. Chem. Soc.* **2017**, *139*, 8698. (e) Ascherl, L.; Evans, E. W.; Hennemann, M.; Di Nuzzo, D.; Hufnagel, A. G.; Beetz, M.; Friend, R. H.; Clark, T.; Bein, T.; Auras, F. Solvatochromic Covalent Organic Frameworks. *Nat. Commun.* **2018**, *9*, 3802.

(7) (a) Sun, Q.; Aguila, B.; Perman, J.; Nguyen, N.; Ma, S. Flexibility Matters: Cooperative Active Sites in Covalent Organic Framework and Threaded Ionic Polymer. *J. Am. Chem. Soc.* **2016**, *138*, 15790. (b) Ding, S.; Gao, J.; Wang, Q.; Zhang, Y.; Song, W.; Su, C.; Wang, W. Construction of Covalent Organic Framework for Catalysis: Pd/COF-LZU1 in Suzuki−Miyaura Coupling Reaction. *J. Am. Chem. Soc.* **2011**, *133*, 19816. (c) Yang, S.; Hu, W.; Zhang, X.; He, P.; Pattengale, B.; Liu, C.; Cendejas, M.; Hermans, I.; Zhang, X.; Zhang, J.; Huang, J. 2D Covalent Organic Frameworks as Intrinsic Photocatalysts for Visible Light-Driven CO_2 Reduction. *J. Am. Chem. Soc.* **2018**, *140*, 14614. (d) Li, H.; Pan, Q.; Ma, Y.; Guan, X.; Xue, M.; Fang, Q.; Yan, Y.; Valtchev, V.; Qiu, S. Three-Dimensional Covalent Organic Frameworks with Dual Linkages for Bifunctional Cascade Catalysis. *J. Am. Chem. Soc.* **2016**, *138*, 14783. (e) Lu, S.; Hu, Y.; Wan, S.; McCaffrey, R.; Jin, Y.; Gu, H.; Zhang, W. Synthesis of Ultrafine and Highly Dispersed Metal Nanoparticles Confined in a Thioether-Containing Covalent Organic Framework and Their Catalytic Applications. *J. Am. Chem. Soc.* **2017**, *139*, 17082.

(8) Davis, M. E. Ordered Porous Materials For Emerging Applications. *Nature* **2002**, *417*, 813.

(9) Wang, X.; Han, X.; Zhang, J.; Wu, X.; Liu, Y.; Cui, Y. Homochiral 2D Porous Covalent Organic Frameworks for Heterogeneous Asymmetric Catalysis. *J. Am. Chem. Soc.* **2016**, *138*, 12332.

(10) Han, X.; Xia, Q.; Huang, J.; Liu, Y.; Tan, C.; Cui, Y. Chiral Covalent Organic Frameworks with High Chemical Stability for Heterogeneous Asymmetric Catalysis. *J. Am. Chem. Soc.* **2017**, *139*, 8693.

(11) (a) Zhang, J.; Han, X.; Wu, X.; Liu, Y.; Cui, Y. Multivariate Chiral Covalent Organic Frameworks with Controlled Crystallinity and Stability for Asymmetric Catalysis. *J. Am. Chem. Soc.* **2017**, *139*, 8277. (b) Xu, H.; Gao, J.; Jiang, D. Stable, Crystalline, Porous, Covalent Organic Frameworks as a Platform for Chiral Organocatalysts. *Nat. Chem.* **2015**, *7*, 905. (c) Xu, H.; Ding, S.; An, W.; Wu, H.; Wang, W. Constructing Crystalline Covalent Organic Frameworks from Chiral Building Blocks. *J. Am. Chem. Soc.* **2016**, *138*, 11489.

(12) (a) Han, X.; Huang, J.; Yuan, C.; Liu, Y.; Cui, Y. Chiral 3D Covalent Organic Frameworks for High Performance Liquid Chromatographic Enantioseparation. *J. Am. Chem. Soc.* **2018**, *140*, 892. (b) Qian, H.-L.; Yang, C.-X.; Yan, X.-P. Bottom-up Synthesis of Chiral Covalent Organic Frameworks and Their Bound Capillaries for Chiral Separation. *Nat. Commun.* **2016**, *7*, 12104.

(13) Friedman, L.; Miller, J. G. Odor Incongruity and Chirality. *Science* **1971**, *172*, 1044.

DOI: 10.1021/jacs.9b02153
J. Am. Chem. Soc. 2019, 141, 7081−7089

(14) Gershenzon, J.; Dudareva, N. The Function of Terpene Natural Products in the Natural World. *Nat. Chem. Biol.* **2007**, *3*, 408.
(15) Pu, L. 1,1'-Binaphthyl Dimers, Oligomers, and Polymers: Molecular Recognition, Asymmetric Catalysis, and New Materials. *Chem. Rev.* **1998**, *98*, 2405.
(16) Pu, L. Enantioselective Fluorescent Sensors: A Tale of BINOL. *Acc. Chem. Res.* **2012**, *45*, 150.
(17) (a) Wanderley, M. M.; Wang, C.; Wu, C.-D.; Lin, W. A Chiral Porous Metal−Organic Framework for Highly Sensitive and Enantioselective Fluorescence Sensing of Amino Alcohols. *J. Am. Chem. Soc.* **2012**, *134*, 9050. (b) Valente, C.; Choi, E.; Belowich, M. E.; Doonan, C. J.; Li, Q.; Gasa, T. B.; Botros, Y. Y.; Yaghi, O. M.; Stoddart, J. F. Metal−Organic Frameworks with Designed Chiral Recognition Sites. *Chem. Commun.* **2010**, *46*, 4911.
(18) Dong, J.; Li, X.; Zhang, K.; Di Yuan, Y.; Wang, Y.; Zhai, L.; Liu, G.; Yuan, D.; Jiang, J.; Zhao, D. Confinement of Aggregation-Induced Emission Molecular Rotors in Ultrathin Two-Dimensional Porous Organic Nanosheets for Enhanced Molecular Recognition. *J. Am. Chem. Soc.* **2018**, *140*, 4035.
(19) Wu, X.; Han, X.; Liu, Y.; Liu, Y.; Cui, Y. Control Interlayer Stacking and Chemical Stability of Two-Dimensional Covalent Organic Frameworks via Steric Tuning. *J. Am. Chem. Soc.* **2018**, *140*, 16124.
(20) Chen, X.; He, Z.; Kausar, F.; Chen, G.; Zhang, Y.; Yuan, W. Z. Aggregation-Induced Dual Emission and Unusual Luminescence Beyond Excimer Emission of Poly(ethylene terephthalate). *Macromolecules* **2018**, *51*, 9035.
(21) (a) Mitra, S.; Kandambeth, S.; Biswal, B. P.; Khayum M, A.; Choudhury, C. K.; Mehta, M.; Kaur, G.; Banerjee, S.; Prabhune, A.; Verma, S.; Roy, S.; Kharul, U. K.; Banerjee, R. Self-Exfoliated Guanidinium-Based Ionic Covalent Organic Nanosheets (iCONs). *J. Am. Chem. Soc.* **2016**, *138*, 2823. (b) Kang, Z.; Peng, Y.; Qian, Y.; Yuan, D.; Addicoat, M. A.; Heine, T.; Hu, Z.; Tee, L.; Guo, Z.; Zhao, D. Mixed Matrix Membranes (MMMs) Comprising Exfoliated 2D Covalent Organic Frameworks (COFs) for Efficient CO_2 Separation. *Chem. Mater.* **2016**, *28*, 1277.
(22) Zhang, Y.; Yuan, S.; Feng, X.; Li, H.; Zhou, J.; Wang, B. Preparation of Nanofibrous Metal−Organic Framework Filters for Efficient Air Pollution Control. *J. Am. Chem. Soc.* **2016**, *138*, 5785.
(23) (a) Dong, J.; Li, X.; Peh, S. B.; Yuan, Y. D.; Wang, Y.; Ji, D.; Peng, S.; Liu, G.; Ying, S.; Yuan, D.; Jiang, J.; Ramakrishna, S.; Zhao, D. Restriction of Molecular Rotors in Ultrathin Two-Dimensional Covalent Organic Framework Nanosheets for Sensing Signal Amplification. *Chem. Mater.* **2019**, *31*, 146. (b) Yan, Z.; Hu, B.; Li, Q.; Zhang, S.; Pang, J.; Wu, C. Facile Synthesis of Covalent Organic Framework Incorporated Electrospun Nanofiber and Application to Pipette Tip Solid Phase Extraction of Sulfonamides in Meat Samples. *J. Chromatogra. A* **2019**, *1584*, 33.
(24) (a) Zhang, Q.; Lei, M.; Kong, F.; Yang, Y. A Water-Stable Homochiral Luminescent MOF Constructed From an Achiral Acylamide-Containing Dicarboxylate Ligand for Enantioselective Sensing of Penicillamine. *Chem. Commun.* **2018**, *54*, 10901. (b) Lee, S. J.; Lin, W. A Chiral Molecular Square with Metallo-Corners for Enantioselective Sensing. *J. Am. Chem. Soc.* **2002**, *124*, 4554. (c) Pugh, V.; Hu, Q.; Pu, L. The First Dendrimer-Based Enantioselective Fluorescent Sensor for the Recognition of Chiral Amino Alcohols. *Angew. Chem., Int. Ed.* **2000**, *39*, 3638. (d) Lin, J.; Hu, Q.; Xu, M.; Pu, L. A Practical Enantioselective Fluorescent Sensor for Mandelic Acid. *J. Am. Chem. Soc.* **2002**, *124*, 2088.
(25) (a) Accetta, A.; Corradini, R.; Marchelli, R. Enantioselective Sensing by Luminescence. *Top. Curr. Chem.* **2010**, *300*, 175. (b) Xuan, W.; Zhang, M.; Liu, Y.; Chen, Z.; Cui, Y. A Chiral Quadruple-Stranded Helicate Cage for Enantioselective Recognition and Separation. *J. Am. Chem. Soc.* **2012**, *134*, 6904.
(26) (a) Lakowicz, J. R. (Ed.). *Principles of Fluorescence Spectroscopy*. 3rd, 2006, Springer. (b) Dong, J.; Tan, C.; Zhang, K.; Liu, Y.; Low, P. J.; Jiang, J.; Cui, Y. Chiral NH-Controlled Supramolecular Metallacycles. *J. Am. Chem. Soc.* **2017**, *139*, 1554.
(27) (a) Ohzono, T.; Yamamoto, T.; Fukuda, J.-i. A Liquid Crystalline Chirality Balance For Vapours. *Nat. Commun.* **2014**, *5*, 3735. (b) Torsi, L.; Farinola, G. M.; Marinelli, F.; Tanese, M. C.; Omar, O. H.; Valli, L.; Babudri, F.; Palmisano, F.; Zambonin, P. G.; Naso, F. A Sensitivity-enhanced Field-effect Chiral Sensor. *Nat. Mater.* **2008**, *7*, 412. (c) Zou, W.; Yan, Y.; Fang, J.; Yang, Y.; Liang, J.; Deng, K.; Yao, J.; Wei, Z. Biomimetic Superhelical Conducting Microfibers with Homochirality for Enantioselective Sensing. *J. Am. Chem. Soc.* **2014**, *136*, 578.

DOI: 10.1021/jacs.9b02153
J. Am. Chem. Soc. 2019, 141, 7081−7089